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Intracardiac delivery catheter and method of use

Abstract

An intracardiac delivery catheter or assembly is disclosed. The delivery catheter includes a delivery lumen and a flow assist balloon inflatable through an inflation lumen. In illustrated embodiments the delivery lumen has a diameter dimension of at least 8 F, 9 F, 10 F or between 8-14 F for insertion of an intracardiac echocardiography catheter (ICE) for placement at an intracardiac imaging site. In embodiments described, the delivery catheter is used for placement of an ICE catheter in a pulmonary artery or other treatment site using an asymmetric balloon having an asymmetric profile and portion to steer the delivery catheter, for example, into the right ventricle and into the pulmonary artery.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority to U.S. Ser. No. 62/861,906 filed Jun. 14, 2019 and entitled INTRACARDIAC IMAGING ASSEMBLY AND METHOD OF USE and U.S. Ser. 63/007,792 filed Apr. 9, 2020 entitled INTRACARDIAC DELIVERY CATHETER AND METHOD OF USE, the content of which are hereby incorporated in the present application in their entirety.

BACKGROUND

(1) Various cardiac treatment procedures require use of an imaging device during the treatment procedure. Intracardiac placement of the imaging or other therapeutic device into the pulmonary artery or other intracardiac treatment site can be difficult and during use, the device can prolapse making the treatment procedure more difficult and cumbersome.

SUMMARY

(2) The present application relates to an intracardiac delivery catheter for intracardiac placement of an imaging or treatment device for use. In the illustrated embodiments, the delivery catheter includes a flow assist balloon for intracardiac placement of the delivery catheter and a delivery lumen therethrough. In illustrated embodiments, the delivery lumen has a diameter dimension of at 8 F, 9 F or 10 F for insertion of an intracardiac echocardiography catheter or intracardiac echocardiography (ICE) catheter and guidewire. For example, in embodiments disclosed, the delivery lumen has a diameter dimension between 8 F-14 F or 9 F-14 F. The delivery lumen

extends from an inlet at the proximal end of a catheter shaft to a distal end of the catheter shaft. In illustrated embodiments an outlet of the delivery lumen is distal from a flow assist balloon of the delivery catheter and is formed through an opened distal end of the catheter shaft and delivery lumen. Embodiments of the delivery catheter include a guide wire lumen in addition to the delivery lumen and an inflation lumen to expand the flow assist balloon for intracardiac placement of the delivery catheter. In an illustrated embodiment, the guide wire lumen is a shortened guide wire lumen.

(3) Embodiments of the intracardiac delivery catheter include an asymmetric shaped flow assist balloon having an asymmetric shape in an inflated or expanded profile. As disclosed the asymmetric shape provides a moment arm or asymmetric portion relative to a longitudinal axis of the catheter shaft to steer the catheter shaft around acute angles, for example through the tricuspid and pulmonary valves for placement in a pulmonary artery. As disclosed, in illustrated embodiments, the delivery lumen is used for intracardiac delivery of materials, imaging catheters or other treatment devices.

(4) Embodiments of the multiple lumen delivery catheter include an inner balloon in the delivery lumen to engage an imaging or echocardiography catheter inserted in the delivery lumen. As disclosed, in illustrated embodiments the length of the delivery catheter is sized for intracardiac placement in a pulmonary artery or other intracardiac site. The delivery catheter can be inserted through a femoral vein into the right atrium through the inferior vena cava or through the superior vena cava via a jugular vein. In illustrated embodiments, the elongate length of the catheter shaft has a varied stiffness or flexibility including a relatively flexible distal portion.

(5) The present application provides a kit for intracardiac imaging including a multiple lumen delivery catheter and intracardiac echocardiography catheter for cardiac imaging and treatment. In illustrated embodiments, the multiple lumens include a delivery lumen, a guide wire lumen and an inflation lumen to inflate a flow assist balloon. In embodiments disclosed the delivery lumen has a diameter of at least 8 F, 9 F or 10 F or between 8 F-14 F or 9 F-14 F for insertion of the intracardiac echocardiography catheter. For use, the delivery catheter is inserted into a right atrium, right ventricle for placement into a pulmonary artery. The delivery catheter is tracked to the right atrium through the inferior vena cava or superior vena cava.

(6) In alternate embodiments, the delivery catheter is used for placement of an imaging catheter in the superior vena cava, pulmonary artery or alternate intracardiac imaging site. In illustrated embodiments, the catheter shaft includes a guidewire lumen for intracardiac placement, a flow assist balloon and a delivery lumen sized for insertion of an intracardiac echocardiography catheter. In another embodiment, the delivery catheter includes a flow assist balloon and a delivery lumen having a smaller dimension for medication or other treatment or imaging devices or asymmetric balloon and delivery lumen. Details of the present invention are described in the following drawings and description.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A illustrates an embodiment of an assembly of the present application.

(2) FIG. 1B is a cross-sectional view as taken along line 1B-1B of FIG. 1A of the flow assisted delivery catheter shown without a guidewire and imaging catheter inserted.

(3) FIG. 1C illustrates a hemostatic connector coupled to a delivery lumen.

(4) FIG. 1D is a detailed cross-sectional view of a portion of the catheter of FIG. 1A with a flow assist balloon illustrated in a deflated condition.

(5) FIG. 1E is a detailed illustration of a distal end of the assembly of FIG. 1A illustrating the flow assist balloon inflated for use.

(6) FIG. 1F schematically illustrates an embodiment of an intracardiac imaging catheter (ICE) of the present application.

(7) FIG. 1G illustrates a proximal end of the assembly and imaging catheter.

(8) FIGS. 2A-2E progressively illustrate a process for using the assembly of the present application for intracardiac imaging.

(9) FIGS. 3A-3B illustrate another embodiment of the assembly of the present application.

(10) FIG. 4A illustrates another embodiment of the assembly of the present application.

(11) FIG. 4B is a cross-sectional view as taken along line 4B-4B of FIG. 4A.

(12) FIG. 4C is a cross-sectional view of a portion of the delivery catheter shown with the imaging catheter in the delivery lumen and an inner balloon inflated.

(13) FIG. 5A illustrates another embodiment of the assembly of the present application.

(14) FIG. 5B is a cross-sectional view as taken along line 5B-5B of FIG. 5A.

(15) FIG. 5C is a sectional view of the distal end of the delivery catheter shown with a guidewire extending through a guidewire lumen formed along a distal end of the delivery lumen.

(16) FIG. 5D is a cross-sectional view as taken along line 5D-5D of FIG. 5C.

(17) FIG. 5E is a sectional view of the delivery catheter having an ICE catheter inserted into the delivery lumen and the inner balloon inflated to abut the ICE catheter.

(18) FIG. 5F is a cross-sectional view as taken along line 5F-5F of FIG. 5E.

(19) FIGS. 6A-6D progressively illustrate process steps illustrating use of the assembly of the present application.

(20) FIG. 6E is a flow chart illustrating steps for cardiac imaging using a delivery catheter of the present application.

(21) FIG. 7A illustrates another embodiment of an intracardiac delivery catheter of the present application.

(22) FIG. 7B illustrates an end view of a flow assist balloon of an embodiment of the intracardiac delivery catheter.

(23) FIG. 7C is a longitudinal cross-sectional view of the intracardiac delivery catheter of FIG. 7A including a stiff proximal length and a relatively flexible distal length.

(24) FIG. 7D is a cross-sectional view taken along line D-D of FIG. 7C.

(25) FIG. 7E is a cross-sectional view taken along line E-E of FIG. 7C.

(26) FIG. 7F is a cross-sectional view taken along line F-F of FIG. 7C.

(27) FIG. 7G is a cross-sectional view taken along line G-G of FIG. 7C.

(28) FIG. 7H is a detailed view of a distal portion of a shaft of the intracardiac delivery catheter of an illustrative embodiment.

(29) FIG. 7I illustrates an embodiment of an asymmetrical balloon at a distal end of the catheter shaft of the intracardiac delivery catheter.

(30) FIG. 7J schematically illustrates a distal end of the catheter shaft of the embodiment shown in FIGS. 7A-7I with an intracardiac imaging catheter (ICE) extending through a distal opening of the catheter shaft.

(31) FIG. 7K illustrates a proximal end of the intracardiac delivery catheter and assembly.

(32) FIG. 8A schematically illustrates force(s) and moment interacting with an asymmetric balloon.

(33) FIG. 8B schematically illustrates an embodiment for intracardiac navigation of an asymmetric balloon.

(34) FIG. 9A schematically illustrates another embodiment of an intracardiac delivery catheter of the present application.

(35) FIG. 9B illustrates an end view of an asymmetric flow assist balloon.

(36) FIG. 9C is a cross-sectional view taken along line C-C of FIG. 9A.

(37) FIG. 9D is a cross-sectional view taken along line D-D of FIG. 9A.

(38) FIG. 10A illustrates another embodiment of an intracardiac delivery catheter of the present application having a shortened guide wire lumen.

(39) FIG. **10B** is an end view of a flow assist balloon of the intracardiac delivery catheter of FIG. **10A**.

(40) FIG. **10C** is a cross-sectional view of the delivery catheter of FIG. **10A** illustrating a shortened guide wire lumen.

(41) FIG. **10D** is a cross-sectional view taken along line D-D of FIG. **10C**.

(42) FIG. **10E** is a cross-sectional view taken along line E-E of FIG. **10C**.

(43) FIG. **11A** illustrates another embodiment of an intracardiac delivery catheter having a shorted imaging catheter (ICE) lumen.

(44) FIG. **11B** is a cross-sectional view of the delivery catheter of FIG. **11A** illustrating the imaging catheter lumen.

(45) FIG. **11C** is a cross-sectional view taken along line C-C of FIG. **11B**.

(46) FIG. **11D** is a cross-sectional view taken along line D-D of FIG. **11B**.

(47) FIG. **12A** illustrates fabrication steps for embodiments of the present application.

(48) FIG. **12B** is a flow chart illustrating a fabrication embodiment of a multiple lumen intracardiac delivery catheter of the present application.

(49) FIG. **13** schematically illustrates an embodiment of a steerable distal tip for embodiments of the delivery catheter of the present application.

(50) In the above-referenced FIGS., like numbers are used to identify like parts. It should be noted that features in the FIGS. are not necessarily drawn to scale.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(51) The present application relates to an assembly and method for intracardiac imaging.

Illustrative embodiments of the assembly use an echocardiography catheter to provide an intracardiac image for medical treatment. Placement of the imaging catheter at an imaging site can be difficult due to the anatomy of the heart or desired image. Embodiments of the present application provide an assembly and method for placement of the imaging catheter in a pulmonary artery, superior vena cava or other intracardiac site to provide a feedback image for treatment.

(52) Illustrative embodiments of the present assembly include an intracardiac delivery catheter (IDC) **100** for placement of the imaging catheter such as an intracardiac echocardiography catheter (ICE catheter) **102** for use. The intracardiac delivery catheter **100** is formed of an elongate shaft **104** having an elongate length extending between a proximal end **106** and a distal end **108**. In an illustrated embodiment, the shaft **104** is formed of a flexible material having sufficient length to track from an insertion site into a pulmonary artery through a patient's right atrium and right ventricle or other treatment as will be described herein. In illustrated embodiments, the insertion site is a femoral vein or jugular vein.

(53) The shaft **104** of the catheter **102** is bendable and has sufficient stiffness and torquability to navigate the circuitous path from the insertion site for intracardiac placement at the imaging site. The length of the catheter body **104** can have varied flexibility to navigate through a patient's body to the intracardiac treatment site. In an illustrated embodiment, the approximate length of the delivery catheter is between 60-90 cm, however in other embodiments, the length is between 50-120 cm. As shown the catheter **102** includes a flow assist balloon **110** coupled to the catheter shaft **104** proximate to the distal end **108**. Inflation fluid is injected into the flow assist balloon **110** through an inflation syringe **112** or other inflation device coupled to the proximal end **106** of the shaft **104**. The proximal end **106** of the shaft **104** also includes a hemostatic connector **116** for the ICE catheter **102** and connector **118** providing a port for a guidewire **120**. In the illustrated embodiment, the hemostatic connector **116** includes a flushing port **122**. As shown, sleeve **124** encloses attachment of the proximal connectors **116**, **118** and syringe **112** to form a proximal hub of the delivery catheter **100**.

(54) As shown in the cross-sectional view of FIG. **1B**, the shaft **104** of the flow assisted delivery catheter **100** includes a plurality of lumens along the length of the shaft **104**. The plurality of lumens include an delivery lumen **130** having a diameter sized for insertion of the ICE catheter **102**

therethrough, a guidewire lumen **132** and an inflation lumen **134**. The delivery catheter lumen **130** has an elongate length that extends from the proximal end **106** of the shaft **104** proximate to the distal end **108**. The ICE catheter **102** is inserted through the hemostatic connector **116** coupled to the proximal end of lumen **130** and is slidably advanced through the lumen **130** for use. In the illustrated embodiment, lumen **130** has an opened distal end. In an illustrated embodiment, a diameter of the delivery lumen **130** is at least 8 F, 9 F or 10 F or between 8-14 F or 9-14 F (French catheter diameter scale) to accommodate a 3D intracardiac echocardiography catheter **102**.

(55) As shown in FIG. 1C, the hemostatic connector **116** is formed of a connector body **140** having a passageway **142** extending between opposed ends of the connector body **140**. The connector body **140** includes a flexible hemostatic valve structure (e.g. gasket) **144** (illustrated diagrammatically) along the passageway **142**. The flushing port **122** is coupled to the passageway **142** of connector **116** through a branch structure **146** having a branch passageway **148**. Flow of fluid through the branch passageway **148** is controlled via a flushing port valve **150** operable between an opened position and a closed position to open and close the flushing port **122**. Application is not limited to a particular hemostatic valve structure **144** and various designs can be employed as will be appreciated by those skilled in the art.

(56) The guidewire lumen **120** has an elongate length that extends from the proximal end **106** of the catheter shaft **104** to the distal end of the catheter shaft **104**. The guidewire **120** is inserted through connector **118** and slidably advanced through lumen **132** for placement through an opened distal end of the guidewire lumen **132**. The inflation lumen **134** extends along the elongate shaft **104** to inject inflation fluid into the flow assist balloon **110** through an inflation port **154** connecting the inflation lumen **134** to an interior of the flow assist balloon **110** as shown in FIG. 1D. The balloon **110** is formed of an elastomeric or other balloon material having a proximal collar **156** connected about the cylindrical body **104** aft of the inflation portion **154** and a distal collar **158** connected about the cylindrical body **104** forward of the inflation port **154**. Inflation fluid is injected into the lumen **134** through syringe **112** to inflate the flow assist delivery balloon **110** as shown in FIG. 1E. As will be appreciated by those skilled in the art, the balloon is a Swan-Ganz type balloon having a construction as described in U.S. Pat. No. 3,634,924, the contents of which is incorporated by reference into the present application.

(57) As previously described, the delivery catheter **100** is configured for placement of ICE catheter **102** at an imaging site via insertion through the delivery lumen **130**. Illustrative imaging catheters include 2D and 3D intracardiac echocardiography imaging catheters. FIG. 1F is an illustrative embodiment of an ICE imaging catheter **102** for placement via delivery catheter **100**. As shown, the ICE catheter **102** has an elongate tubular body or shaft **160** having an elongate length extending between a proximal end **162** and a distal end **164**. The elongate length of the tubular body has sufficient flexibility, steerability and length for intracardiac placement in a pulmonary artery or other intracardiac imaging site from an insertion site. A plurality of transducer elements or transducer arrays **166** (illustrated diagrammatically) are coupled to the distal end of the elongate body **160** to generate ultrasonic waves or vibration and detect feedback vibration or excitation for imaging. As shown the transducer elements **166** are coupled to a control unit **170** through a plurality of wires/leads **172** and socket connection **180** to provide an input excitation or vibration and receive feedback vibration or excitation. In an illustrated embodiment, the ICE catheter **102** includes separate transducers **166** for transmitting and receiving ultrasonic vibration or the same transducers can be used as both transmitters and receivers.

(58) The control unit **170** includes circuitry to generate ultrasonic or echoing high frequency vibration in response to input control **174** and process feedback signals from the transducer elements **166** to provide an output image **176** for display on a GUI device **178**. The control unit **170** is coupled to a power source for use and is coupled to the transducer elements **166** through one or more circuit elements and/or flex circuits. Illustrative transducer elements **166** include an array of piezoelectric elements, capacitive elements or other electromechanical transducer elements for

generating ultrasonic vibration or disturbance and detecting feedback vibrations. In an illustrated embodiment as shown in FIGS. 1E-1F, the imaging catheter **102** has a rounded or contoured distal tip for intracardiac placement. As shown in FIG. 1G, in an illustrated embodiment, the plug and socket connection **180** includes a plug **180a** connected to the distal end of the ICE catheter **102** which connects to a socket (not shown) connected to the control unit **170**.

(59) As previously described, the present delivery catheter **100** has application for intracardiac placement of the ICE catheter **102** in the heart. The heart includes a right ventricle (RV) which provides blood flow into the pulmonary artery (PA). Blood flows into the right ventricle (RV) from the right atrium (RA) and from the superior vena cava or the inferior vena cava. The pulmonary artery (PA) is near the aorta which receives blood flow from the right ventricle (RV). The position of the pulmonary artery (PA) near the aorta provides a practical site to image the aorta or other structures of the heart for both electrophysiology and structural procedures. Because of the structure of the heart, placement of the ICE catheter **102** through the right atrium (RA) into the right ventricle (RV) and pulmonary artery (PA) is difficult. In particular, in order to position the imaging catheter **102** into the pulmonary artery (PA) from the right ventricle (RV) the catheter must be guided and steered along an “S” curve path and often can fall back into the right ventricle due to the tortuous path. Use of the present assembly facilitates placement of an ICE catheter **102** as progressively illustrated in FIGS. 2A-2E

(60) As shown, in FIG. 2A, the delivery catheter **100** is inserted into the right atrium (RA) for example through the inferior vena cava via a femoral vein or the superior vena cava (not shown) through a jugular vein. As shown in FIG. 2B, inflation fluid is injected into the flow assist balloon **110** via inflation lumen **134** to inflate the flow assist balloon **110**. The balloon **110** is inflated to take advantage of the blood flow through into and through the right atrium (RA) into the right ventricle (RV) and pulmonary artery (PA). In particular, inflation of the flow assist balloon **110** provides an expanded profile to interface with the blood flow from the right atrium (RA) into the right ventricle (RV) and pulmonary artery (PA) as illustrated by arrows **182**. The interface with the blood flow provides an input force or momentum that assists movement or rotation of the delivery catheter **100** from the right atrium (RA) into the right ventricle (RV) and pulmonary artery (PA). In an illustrative embodiment, the balloon is inflated prior to insertion through the tricuspid valve and pulmonary valve to protect the valves from damage.

(61) As shown in FIG. 2C, a guidewire **120** is inserted through the guidewire lumen **132** of the delivery catheter **100**. The guidewire **120** has sufficient stiffness and rigidity to straighten out the curve from the right atrium (RA) to the pulmonary artery (PA) for preparation for insertion of the intracardiac imaging catheter **102** as shown in FIG. 2D. In particular, the guidewire **120** reduces an “S” curve configuration to provide a straighter path for placement of the ICE catheter **102**. As shown in FIG. 2D, the intracardiac imaging catheter **102** is inserted through lumen **130** of the delivery catheter **100** for placement in the pulmonary artery (PA) for imaging. In illustrative embodiments, the guidewire **120** is inserted with the delivery catheter **100** or subsequent to intracardiac placement of the delivery catheter **100**. In particular, the delivery catheter **100** can be inserted with the guidewire **120** for support.

(62) As shown in FIG. 2E, once the imaging catheter **102** is in place, the delivery catheter **100** and guidewire **120** are withdrawn. In illustrative embodiments, the guide wire **120** is useful to re-advanced the delivery catheter **100** or other catheter into the pulmonary artery from the right ventricle (RV) or superior vena cava from the right atrium for treatment or imaging and is retained in the guidewire lumen **132**. In alternate embodiments, the guidewire **120** is withdrawn as shown in FIG. 2E, but the delivery catheter **100** remains in place in the pulmonary artery (PA) during imaging or both the delivery catheter **100** and guidewire **120** remain in place.

(63) In particular, in an illustrated embodiment, the imaging catheter **102** and transducer elements **166** are disposed in the delivery lumen **130** during imaging as described in illustrated embodiments herein. A transmission fluid is inserted into the delivery lumen through port **122** to provide a

transmission medium for imaging. In alternate embodiments, the length of the delivery catheter **100** is sized so that the ICE or imaging catheter **102** extends outside the delivery catheter lumen **130** during imaging. In illustrative embodiments, the delivery catheter **100** remains in place to prevent prolapse of the ICE catheter **102** and facilitate torquability of the ICE catheter **102**. The delivery catheter **100** can prevent prolapse of the imaging ICE catheter **102** from the (PA) into the (RV) and from the superior vena cava (SVC) into (RA).

(64) Thus, as described, the delivery catheter **100** facilitates intracardiac placement of ICE catheter **102**. In one embodiment, the ICE catheter **102** is positioned in the mid right atrium (RA) and the transducer array **166** or transducers are orientated toward the tricuspid valve to image the right ventricle, a portion of the right atrium and the tricuspid valve. In another embodiment, the ICE catheter **102** is orientated to provide an image of the aortic valve, coronary sinus and atrial septum for a transseptal procedure. The ICE catheter **102** can be placed in the right ventricle and orientated to image the left ventricle to screen for effusion or thrombus or mitral valve anatomy. Use of the delivery catheter **100** with the ICE catheter **102** enhances torquability of the ICE catheter **102** within the delivery catheter **100**. The delivery catheter may also be positioned across the atrial septum into a pulmonary vein to support an ICE probe to examine the mitral valve or left atrial appendage.

(65) FIGS. 3A-3B illustrate an alternate embodiment of an assembly of the present application including a flow assisted delivery catheter **100** for use in combination with an imaging or ICE catheter **102** where like numbers are used to refer to like parts in the previous FIGS. As shown in FIGS. 3A-3B, the shaft **104** includes a delivery lumen **130** sized for insertion of the ICE or imaging catheter **102** and guidewire **120** therethrough. For use, the guidewire **120** and ICE catheter **102** are inserted through connector **116** into the delivery lumen **130** for placement of the wire **120** and ICE catheter **102** for treatment as previously described and illustrated in FIGS. 2A-2E. The flow assist balloon **110** is inflated via fluid injected into the inflation lumen **134** via syringe **112** through inflation port **154** as described in the previous embodiment. In an alternate embodiment not shown, the delivery lumen **130** is sized to accommodate insertion of the ICE catheter **102** and guidewire **120** individually and the guidewire **120** is withdrawn prior to insertion of the ICE catheter **102** as described in illustrative embodiments.

(66) FIGS. 4A-4B illustrate another embodiment of the delivery catheter **100** including a flow assist balloon **110** coupled to the distal end of the delivery catheter **100** and plurality of lumens along the body **104** as shown in FIG. 4B. The plurality of lumens include a delivery lumen **130** which is sized to accommodate the guidewire **120** and ICE catheter as shown in FIG. 4A and inflation lumen **134** to inflate the flow assist delivery balloon **110**. The flow assist balloon **110** is inflated via inflation fluid injected into inflation lumen **134** through syringe **112A** as shown in FIG. 4A. The delivery catheter **100** as shown in FIG. 4B also includes an inflation lumen **190** to inflate an inner balloon **192** disposed in the distal end of the delivery lumen **130** as shown in FIG. 4C. Inflation fluid is injected through the lumen **190** into the balloon **192** through inlet port **194**.

(67) The inner balloon **192** is inflated to retain the ICE catheter for use and provide an ultrasonic transmission fluid or medium through the inner balloon **192** as shown in FIG. 4C. In particular, the inner balloon **192** includes a proximal collar **196** coupled to the inner surface of the delivery lumen **130** aft of the inlet port **194** and a distal collar **198** coupled to the delivery lumen **130** forward of the inlet port **194**. Inflation fluid is injected into lumen **190** via syringe **112B** to inflate the inner balloon **192** to engage the ICE catheter disposed in the delivery lumen **130** for use. In an alternate embodiment not shown, the embodiment shown in FIGS. 4A-4B with the inner balloon **192** includes a separate guidewire lumen **132**.

(68) FIGS. 5A-5F illustrate another embodiment of an assembly of the present application. As shown, the assembly includes a delivery catheter **100** having an elongate shaft **104** and elongate guidewire **120**. The delivery catheter **100** includes a flow assist balloon **110** coupled to the distal end of the elongate shaft **104** for intracardiac placement as previously described. The elongate shaft

104 includes a plurality of lumens including a delivery lumen **130** for an ICE catheter, inflation lumen **134** for inflating the flow assist balloon **110** and an inner inflation lumen **190** to inflate an inner balloon **192**. Fluid is supplied into the inflation lumen **134** via syringe **112A** and into inflation lumen **190** through syringe **112B**.

(69) As shown in FIGS. 5A and 5C the delivery catheter includes a rapid exchange guidewire lumen **200** at the distal end of the delivery catheter **100** for the guidewire **120**. In the illustrated embodiment, the rapid exchange guidewire lumen **200** is formed along a distal length of the delivery lumen **130**. The rapid exchange guidewire lumen **200** includes a proximal opening **202** formed through the shaft **104** into the delivery lumen **130** and a distal opening **204** formed through an opened distal end of the delivery lumen **130**. The distal end of the wire **120** extends through the distal opening **204** along the delivery lumen **130** (guidewire lumen **200**) and through the proximal opening **202** to track the delivery catheter **100** to the treatment site as shown in FIGS. 5C-5D.

(70) As shown in FIG. 5E, the delivery catheter **100** also includes inner balloon **192** along the delivery lumen **130**. The inner balloon **192** is inflated to engage the ICE catheter **102** inserted through delivery lumen **130** as shown in FIG. 5E. Fluid is injected into the inner balloon **192** through the inner inflation lumen **190** to inflate the inner balloon **192** via syringe **112B** to abut the ICE catheter **102**. As shown the inner balloon **192** includes proximal collar **196** coupled to inner surface of the delivery lumen **130** aft of the inlet port **194** and distal collar **196** coupled to an inner surface of the delivery lumen **130** spaced forward of the inlet **194**. The inflated inner balloon **192** stabilizes the ICE catheter **102** and provides a transmission fluid surrounding the ICE catheter and in particular, the transducer elements **166** of the ICE catheter.

(71) While in the illustrated embodiments, the rapid exchange or shortened guidewire lumen **200** is formed along a distal length of the delivery lumen **130**, application is not limited to the illustrated embodiment and the rapid exchange guidewire lumen **200** can be formed of a separate shortened guidewire lumen at the distal end of the delivery catheter **100**. Further the delivery catheter **100** can include a plurality of openings **202** for the guide wire **120** spaced about the circumference of the delivery catheter **100**. In an illustrative embodiment, opening **202** is spaced 1-3 centimeters from the distal tip or end **108** of the delivery catheter **100**.

(72) FIGS. 6A-6D illustrate use of delivery catheter **100** having a rapid exchange or shortened guidewire lumen **200** as described. As shown, in FIG. 6A, in the illustrated embodiment, the guidewire **120** is percutaneously inserted into the patient and guided for intracardiac placement. In particular, in the illustrated embodiment, the guidewire **120** is advanced intravascularly into the right atrium to facilitate insertion of the delivery catheter **100**. As shown, in FIG. 6B, the delivery catheter **100** is advanced along the guidewire **120** for intracardiac placement. As shown in FIG. 6C, inflation fluid is injected through syringe **112A** to inflate balloon **110** to provide flow assistance or force to track the delivery catheter **100** into the right ventricle and the pulmonary artery to a treatment or imaging site. In particular, the balloon **110** is inflated to use blood flow to track the delivery catheter **100** into the right atrium, right ventricle and pulmonary artery.

(73) In FIG. 6D when the delivery catheter **100** is in place, the balloon **110** is deflated and the ICE catheter **102** is inserted. In illustrative embodiments, the balloon **110** is deflated and the guidewire **120** is withdrawn prior to placement of the ICE catheter **102** or in illustrated embodiments the guidewire is withdrawn prior to inflation of the flow assist balloon **110**. In other embodiments, the guidewire remains in place for insertion of the imaging catheter **102** and during imaging. As previously described, the delivery catheter **100** can be withdrawn or remain in place during imaging.

(74) FIG. 6E illustrates steps of the illustrated assemblies of the present application. As shown in step **210**, a guidewire **120** is inserted and tracked intravascularly for intracardiac placement, for example to insert the delivery catheter **100** into the right atrium. In step **212**, the delivery catheter **100** is inserted for intracardiac placement using the guidewire **120**. In step **214** the flow assist balloon **110** on the delivery catheter **110** is inflated to assist placement of the delivery catheter **100**

at the intracardiac treatment site, for example into the right atrium, right ventricle or pulmonary artery. In step **216** the ICE or imaging catheter **102** is advanced through the delivery lumen **130** to an imaging or treatment site for example, in the pulmonary artery. In an example embodiment, the flow assist balloon **110** is deflated prior to insertion of the ICE catheter **102**. The inner balloon **190** is inflated in step **218** to abut the ICE catheter **102** and provide a fluid bladder for transmission of ultrasonic imaging waves during use. Alternatively, fluid is injected into the delivery lumen for imaging. In illustrative embodiments, the guidewire is withdrawn prior to insertion of the ICE catheter or in an alternate embodiment the guidewire **120** is withdrawn prior to inflation of the flow assist balloon. In illustrative embodiments, the delivery catheter **100** includes a separate guidewire lumen **132**, **200** and the guidewire **120** remains in place to reposition the delivery catheter **100** for use.

(75) In an alternate embodiment of a delivery catheter **100** having an inner balloon **192** and outer balloon **110**, the delivery catheter **100** includes delivery lumen **130**, inflation lumen **134**, inflation lumen **190** and a guide wire lumen **120** that extends along the length of the delivery catheter between the proximal and distal ends.

(76) FIG. 7A illustrates another embodiment of an intracardiac echocardiography delivery catheter **100A** having an elongate cylindrical shaped body or shaft **104A** where like numbers are used to identify like parts in the previous FIGS. As shown, the elongate body or shaft **104A** includes a relatively stiff proximal length **220** and a relatively or ultra-flexible distal length **222**. The relatively stiff proximal length **220** provides pushability and torquability for insertion and the relatively flexible length **222** provides sufficient flexibility or bendability for intracardiac navigation, for example, through the inferior vena cava into the right atrium, right ventricle and into the pulmonary artery for placement of an imaging or ICE catheter **102**. As previously described, placement of the delivery catheter **100A** is assisted by a flow assist balloon **110A** at the distal end of the catheter shaft **104A** (or ultra-flexible distal length **222**).

(77) In the illustrated embodiment, the balloon **110A** is an asymmetrical balloon as shown in FIG. 7B to enhance navigation and placement of the catheter **100A** as will be described herein. Balloon **110A** is inflated via syringe **112** to assist placement of the delivery catheter **100A** as previously described. In an illustrative embodiment, a length dimension of the relatively stiff proximal length **220** of the catheter is approximately 30-50 cm (11-20 inches) and a length dimensions of the flexible distal length **222** of the catheter is approximately, 30 cm (11 inches).

(78) In the illustrated embodiment shown in FIG. 7A, the catheter shaft **104A** of the delivery catheter **100A** is a multiple lumen shaft **104A** as shown in FIGS. 7C-7G. As shown, the multiple lumen shaft **104A** includes an ICE delivery lumen **130**, guidewire lumen **132**, and a balloon inflation lumen **134** formed along the proximal and distal lengths **220**, **222** of the shaft **104A**. In the embodiment shown, the multiple lumens are formed along an inner core or lumen structure formed of a plurality of microtubes **230** for each of the plurality of lumens. In illustrated embodiments, the plurality of microtubes **230** are formed of polytetrafluoroethylene (PTFE), such as PTFE sold under the trademark Teflon® by Chemours Company FC of Delaware. The Teflon® microtubes **230** provide a lubricous inner surface for use and for slideable placement and movement of the ICE catheter **102** and guide wire **120**.

(79) The stiff proximal length **220** is formed of a stiff perimeter or wall structure **232** formed about the core or lumen structure. In illustrated embodiments, the stiff perimeter structure **232** of the proximal length **220** is formed of hard durometer polymer material such as a polyether block amide, for example, Pebax® available from Arkema France Corporation of Colombes France having a hardness range of 63 Shore D, or polyvinyl alcohol (PVAL) urethane material or similar material having a hardness range between 25-72 Shore D to form the relatively stiff proximal length **220** of the catheter shaft **104A**.

(80) The flexible distal length **222** of the catheter shaft **104A** has a relatively flexible perimeter or wall structure **234** formed of helical coil **240** embedded within a soft polymer body or layers **242**

(inner and outer layers). The helical coil **240** is formed of an aluminum, stainless steel or other material. In an illustrated embodiment the helical coil **240** provides a relatively high hoop strength to provide structural integrity for the multiple lumens of the catheter shaft **104A** and the polymer body or layers **242** and embedded helical coil **240** provide a flexible and bendable distal length for intracardiac placement of the catheter **100A**. Illustratively, flexible polymer body or layers **242** are formed of a low durometer or hardness material to provide the flexibility for the distal length **222**. Illustrative low durometer materials include polychloroprene layer(s) or material having a hardness of approximately 60 Shore A or other material having similar durometer or hardness.

(81) As shown in FIG. 7C the catheter shaft **104A** includes a transition structure connecting the proximal and distal lengths **220**, **222** to form the elongate length of the catheter shaft **104A**. As shown, the transition structure includes a tubular braid **252** having a length that extends along a transition zone between the proximal and distal lengths **220**, **222** to connect the proximal and distal lengths of the catheter shaft **104A**. The tubular braid **252** encloses the core or lumen structure as shown FIGS. 7C and 7F and is formed of a Nitinol-nickel-titanium alloy, stainless steel or fabric material. As shown, the relatively stiff perimeter structure **232** of the proximal length **220** and flexible perimeter or wall structure **234** of the distal length **222** are formed along the proximal and distal ends of the transition zone, respectively. In addition to the structure shown, in illustrated embodiments, the catheter shaft **104A** includes on or more stiffening wires formed along the structure of the catheter shaft **104A** for steerability.

(82) In the illustrated embodiment shown, the core or lumen structure is formed of a plurality of microtubes **230** that extend along the length of the catheter shaft **104A**, however, in alternate embodiments, the microtubes **230** extend along the proximal length of the catheter shaft and the lumens along the distal length are formed or molded along the distal perimeter structure **234** or shaft **104A**. As previously described, and as shown in FIG. 7H, the length of the inflation lumen **134** extends from the proximal end of the catheter shaft **104A** to a distal port **154** opened to an interior of the balloon **110A** to inflate the balloon **110A** at the distal end **108** of the catheter shaft **104A** as shown in FIGS. 7I-7J. The guide wire lumen **132** and ICE delivery lumen **130** extend the length of the catheter shaft **104A** from the proximal end **106** to distal openings at the end of the catheter shaft **104A**. FIG. 7K illustrates the syringe **112** (with control valve), ICE catheter **102** and guide wire connector **118** or port at the proximal end of the catheter shaft **104A**.

(83) As previously described and shown in FIG. 7B, in an illustrative embodiment, the balloon **110A** is an asymmetric balloon where centroid c of the body of the balloon **110A** is off-set from a longitudinal axis of the catheter shaft $C_{sub.s}$ as shown in FIG. 8A. During use as schematically shown in FIG. 8A, blood flow or cardiac pressure provides a distributed force along a surface of the balloon **110A**. In contrast to a symmetrical balloon, force applied to the asymmetrical balloon **110A** imparts a resultant force $F_{sub.r}$ through a centroid c of the balloon spaced from the longitudinal axis of the catheter shaft $C_{sub.s}$ to define a moment arm r or asymmetric portion between the catheter shaft $C_{sub.s}$ and centroid c . The resultant force $F_{sub.r}$ and moment arm r imparts a moment $M_{sub.CS}$ relative to the catheter shaft $C_{sub.s}$ having a magnitude provided by $M_{sub.CS} = F_{sub.r} * r$. Thus, as described, the asymmetric portion of the balloon **100A** is orientated to assist with steering and placement of the delivery catheter **100** for use. For example, the asymmetric portion is oriented to steer the catheter **100** through the tricuspid valve into the right ventricle and through the pulmonary valve into the pulmonary artery.

(84) As shown in FIG. 8B during use and insertion, the balloon is orientated to impart a clockwise moment $M_{sub.CS}$ to align a tip of the delivery catheter **100A** with the tricuspid valve for insertion into the (RV) and as shown in FIG. 8B, the balloon **110A** is orientated to impart a counterclockwise moment $M_{sub.CS}$ relative to the catheter shaft **104A** to align the tip of the delivery catheter **100A** with the pulmonary artery (PA) for access to an imaging or treatment site. Additionally, the off-axis alignment of the balloon reduces the surface area to reduce back pressure and force imparted to the catheter **100A** during insertion.

(85) While illustrative embodiments show application of the asymmetric balloon for placement of the delivery catheter **100A**, application of the asymmetric balloon **110A** as described is not limited to delivery catheter **100A** and can be used for navigation of other intracardiac or delivery catheters into the body or to a treatment site. In an illustrative embodiment, the asymmetric balloon **110A** is formed of an elongate generally tubular shaped body having spaced first and second ends connected to an outer surface of the flexible perimeter or wall structure **234** or catheter shaft **104A**. The asymmetric shape is formed via a thinner or stressed wall thickness of the tubular shaped body so that inflation of the interior cavity of the balloon forms the asymmetric shape of the balloon **110A**.

(86) FIGS. **9A-9D** illustrate an alternate embodiment of the delivery catheter **100A** having a stiff proximal length **220** and are relatively flexible distal length **222** as previously described with respect to FIGS. **7A-7G**. In the embodiment shown, the multiple lumen catheter shaft **104A** includes ICE delivery lumen **130** and balloon inflation lumen **134**. Similar to the embodiment illustrated in FIGS. **3A-3B**, the delivery lumen **130** is sized to accommodate the ICE imaging catheter **102** and guidewire **120** for placement of the delivery catheter **100A** for use.

(87) FIGS. **10A-10E** illustrate another embodiment of a delivery catheter **100A** including a stiff proximal length **220** and a flexible distal length **222** as previously described. In the illustrated embodiment shown, the shaft **104A** includes an ICE delivery lumen **130** and an inflation lumen **134** along the proximal and distal lengths **220**, **222**. As shown in FIG. **10C**, the distal length includes a shortened guidewire lumen **132** (**200**). The rapid exchange or shortened guidewire lumen **132** (**200**) includes a proximal opening **202A** and a distal opening **204A** (as shown in FIG. **10A**) formed through an opened distal end of lumen **132** at the distal end of the catheter shaft **104A**. The distal end of the wire **120** extends through the proximal opening **202A** and the distal opening **204A** to track the delivery catheter to the treatment site. The proximal opening **202A** is spaced from the proximal end of the catheter shaft **104A**. In the embodiment shown, the length of the rapid exchange or shortened guidewire lumen **132** (**200**) extends along the flexible distal length **222**, however, application is not limited to the length of the rapid exchange guide wire lumen **132** shown and in illustrative embodiments the length can extend along the entire flexible distal length **222** or along the distal length and a portion of the stiff proximal length **220** so that the proximal opening **202A** is located along the stiff proximal length **220**.

(88) FIGS. **11A-11D** illustrate another embodiment of a delivery catheter **100A** including a stiff proximal length **220** and a flexible distal length **222** as previously described. In the illustrated embodiment, the catheter **100A** includes a shortened ICE delivery lumen **130** along the relatively flexible distal length **222** of the catheter shaft as shown in FIG. **11B**. As shown, the flexible distal length **222** includes ICE delivery lumen **130** having a proximal opening **202B** spaced from the proximal end **106** of the catheter shaft **104A** and a distal opening **204B** (shown in FIG. **11A**) at the distal end **108** of the catheter shaft **104A**.

(89) Thus, as shown, the ICE delivery lumen **130** is formed along the flexible distal length **222** of the catheter shaft. For use, the stiff proximal length **220** provides sufficient rigidity and pushability to insert the delivery catheter **100A** and ICE catheter **102** through a patient's vascular to the right atrium. Guide wire **120** is inserted through lumen **132** and the balloon **110A** is inflated to float the delivery catheter into right ventricle (RV) and pulmonary artery (PA) for imaging. In alternate embodiment, not shown, both the guide wire lumen **132** and ICE delivery lumen **130** have a shortened length extending along the flexible distal length **222** of the catheter shaft **104A**. In an alternate embodiment not shown, the shortened ICE delivery lumen **130** extends along the flexible distal length **222** and a portion of the stiff proximal length **220** so that proximal opening **202B** is disposed along the stiff proximal length **220** of the catheter shaft **104A**.

(90) FIG. **12A** schematically illustrates fabrication of a multiple lumen delivery catheter **100A** having a stiff proximal length **220** and an ultra-flexible distal length **222** in accordance with illustrative embodiments described. As progressively shown in step **300**, a plurality of mandrels

302A, 302B, 302C having different sized diameters to correspond to the different sized lumens are coated with a Teflon® coating or other lubricous material. In step **302**, microtubes **230** are assembled on the mandrels **302A-302C** to form the plurality of lumens. As previously discussed, in illustrated embodiments, the microtubes **230** are formed of a Teflon material to provide a lubricous inner surface of the lumens. In step **304**, the microtubes **230** are joined along a length thereof to form a multiple lumen or core structure. Illustratively, the microtubes **230** are connected via an adhesive or other material. An outer surface of the tubular liners can be roughened to enhance the adhesive connection of the microtubes **230**.

(91) As shown in step **306**, a tubular braid **252** is deposited over an outer circumference of the microtubes to form the transition between the proximal and distal lengths **220, 222** of the catheter shaft. In step **308**, an outer surface of the core or lumen structure is coated with the hard polymer material along proximal length and tubular braid **252** along a transition length and the soft polymer material or layers **242** and helical coil **240** are deposited over a distal length of the core or lumen structure and a transition length of the tubular braid **252** to form the proximal and distal lengths **220, 222** of the catheter shaft **104A**. As shown in step **310**, a heat shrink tube **311** is placed about the length of the structure and in step **312**, heat is supplied to shrink tube **311** to impart pressure to the polymer coatings or layers. In an illustrated embodiment the heat shrink tube is a fluorinated ethylene propylene (FEP) heat shrink tube. Heat and pressure imparted by the shrinking diameter of the shrink tube **311** causes the polymer coatings or layers to flow into the interspaces of the tubular braid **252** to connect the relatively stiff proximal length **220** of the catheter and the relatively flexible distal length **222** of the catheter and coil. Thereafter, in step **314** the heat shrink tube **311** and mandrels **302** are removed for assembly of the proximal hub and balloon to the catheter shaft **104** to form the delivery catheter **100A**.

(92) In alternate embodiments, the length of the microtubes **230** extends along the proximal length **220** and not along the distal length **222** and the polymer coating or layers form the lumen structure for the multiple lumens along the distal length **222**. In alternate embodiments as disclosed shortened microtubes **230** are used to form a shortened distal guide wire lumen **132** or ICE delivery lumen **130**.

(93) FIG. **12B** is a flow chart illustrating process steps for fabrication of the delivery catheter **100A** and shaft according to embodiments of the present application. As previously described a plurality of mandrels **302** are used to form a plurality of lumens of a multiple lumen catheter shaft. As shown in step **350** a plurality of microtubes **230** are assembled over mandrels and adhesively connected to form the multiple lumen/core structure. In step **352**, tubular braid **252** is assembled over the lumen or core structure to form a transition zone. In step **354** a proximal length of the structure is coated with a hard durometer polymer material to form the stiff proximal length **220** of the catheter. In step **356**, a helical spring is assembled over the distal length between inner and outer low durometer polymer layers to form the flexible distal length **222**. In step **358** heat is applied to a heat shrink tube **311** to provide heat and pressure about an outer circumference of the coatings or layers to provide material flow to form the perimeter wall structures of the proximal and distal lengths **220, 222** of the catheter shaft connected along a transition length through the braided tube **252**. In step **360** the heat shrink tube **311** and mandrels **302** are removed.

(94) FIG. **13** illustrates a distal tip of an illustrative embodiments of the delivery catheter **100, 100A**. As shown, elongate control wires **340** are coupled to the distal tip of the delivery catheter **100, 100A** to adjust an angle (as shown in phantom) of the distal tip of the delivery catheter relative to the longitudinal length of the shaft **104, 104A**. As schematically shown, the length of the control wire **340** extends from the distal tip to the proximal end of the catheter shaft **104** or hub so that the user can intravenously adjust the angle of the distal tip to steer the delivery catheter **100, 100A** to the treatment or imaging site via movement of the control wire(s) **340** as illustrated by arrows **342**. In illustrated embodiments, one or more control wires **340** extend along one or more control wire lumens (not shown) formed along the catheter shaft **104** and illustrative embodiments include one

or more control wires **340** and one or more control wire lumens (not shown).

(95) While illustrative embodiments are shown, application of the present invention is not limited to the illustrated embodiments and changes and modifications can be made as will be appreciated by those skilled in the art. For example, the flow assisted delivery catheter **100** can be used for placement of an ICE or imaging catheter **102** at alternate imaging sites. Further application of the present invention is not limited to the particular order of steps described. As described, application of the illustrative embodiments allow for placement of an imaging catheter at an intracardiac imaging site without damage to the superior vena cava or placement in a pulmonary artery or other intracardiac treatment site without damage to tricuspid or pulmonary valves via a balloon tipped delivery catheter **100**. While specific combination of elements are shown in illustrated embodiments, it should be appreciated that features and elements of the illustrated embodiments can be combined to form alternate embodiments of the invention.

Claims

1. A method for providing an echocardiographic image comprising steps of: intravascularly inserting an intracardiac delivery catheter including an asymmetric shape flow assist balloon at a distal end of the delivery catheter for placement of an imaging catheter including one or more transducer elements; inflating the asymmetric shape flow assist balloon at the distal end of the delivery catheter; using the inflated flow assist balloon to advance the intracardiac delivery catheter; orientating an asymmetric portion of the asymmetric shape flow assist balloon in a first orientation to form a moment arm to impart rotation to the delivery catheter to align a tip of the delivery catheter with a right ventricle and advancing the delivery catheter through a tricuspid valve into the right ventricle; orientating the asymmetric portion of the asymmetric shape flow assist balloon in a second orientation different from the first orientation to align the moment arm to impart rotation to the delivery catheter to align the tip of the delivery catheter with a pulmonary artery and advancing the delivery catheter through a pulmonary valve into the pulmonary artery; inserting the imaging catheter into a delivery lumen of the delivery catheter; slideably advancing the imaging catheter through the delivery lumen of the delivery catheter to position the imaging catheter at an intracardiac treatment site in the pulmonary artery; and receiving image feedback from the one or more transducer elements of the imaging catheter.
2. The method of claim 1 wherein the steps of orientating the asymmetric portion of the flow assist balloon include: rotating a shaft of the delivery catheter to a first position to locate the flow assist balloon in the first orientation; and rotating the shaft of the delivery catheter to a second position to locate the flow assist balloon in the second orientation.
3. A method for providing an echocardiographic image comprising the steps of: intravascularly inserting an intracardiac delivery catheter including an asymmetric shape flow assist balloon at a distal end of the delivery catheter for placement of an imaging catheter including one or more transducer elements; inflating the asymmetric shape flow assist balloon at the distal end of the delivery catheter; using the inflated flow assist balloon to advance the intracardiac delivery catheter; orientating an asymmetric portion of the asymmetric shape flow assist balloon to form a moment arm to impart rotation to a tip of the delivery catheter to align the tip of the delivery catheter with a pulmonary artery; advancing the delivery catheter through a pulmonary valve into the pulmonary artery; inserting the imaging catheter into a delivery lumen of the delivery catheter; slideably advancing the imaging catheter through the delivery lumen of the delivery catheter to position the imaging catheter at an intracardiac treatment site in the pulmonary artery; and receiving image feedback from the one or more transducer elements of the imaging catheter.
4. The method of claim 3 wherein the step of orientating the asymmetric portion of the flow assist

balloon comprises: rotating a shaft of the delivery catheter to align the tip of the delivery catheter for insertion into the pulmonary artery.
